

Special Relativity Equation Cheat Sheet

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github.com/zettix/SpecialRelativityEquationsCheatSheet

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1 Introduction

This is a collection of equations to reference when learning SR on Youtube. Stanford has a class available here: Special Relativity, Stanford. Taught by Leonard Susskind. <https://www.youtube.com/watch?v=Z8ace3NbXAA&list=PLD9DDFBDC338226CA>

2 The Equations

Speed of light

$$\begin{aligned}c &\approx 300,000,000 \text{ m/s} = 3 \times 10^8 \text{ m/s} \\ &= 299,792,458 \text{ m/s}\end{aligned}\tag{1}$$

Proper Time (Invariant)

$$\tau^2 = t^2 - x^2\tag{2}$$

$$\Delta\tau^2 = \Delta t^2 - \Delta x^2\tag{3}$$

Lorentz Transformations

$$t' = \frac{t - vx}{\sqrt{1 - v^2}}\tag{4}$$

$$t = \frac{t' - vx'}{\sqrt{1 - v^2}}\tag{5}$$

$$x' = \frac{x - vt}{\sqrt{1 - v^2}}\tag{6}$$

$$x = \frac{x' - vt'}{\sqrt{1 - v^2}}\tag{7}$$

Newtonian Momentum

$$p = mv \quad (8)$$

Newtonian Force

$$f = ma = mv^2 = pv \quad (9)$$

4-Space: with time, X has μ superscript.

$$X^\mu = (t, x, y, z) \quad (10)$$

4-Space: without time, X has i superscript.

$$X^i = (x, y, z) \quad (11)$$

4-Dimensional Velocity

$$\frac{\Delta X^\mu}{\Delta \tau} \quad (12)$$

4-Dimensional Time Velocity

$$U_0 = U^0 = \frac{1}{\sqrt{1 - v^2}} \quad (13)$$

4-Dimensional Space Velocity

$$\vec{U} = U^i = U_i = \frac{v^i}{\sqrt{1 - v^2}} \quad (14)$$

Note

$$U_0^2 - \vec{U}^2 = 1 \quad (15)$$

Action Potential Sums

$$A = -m \sum \Delta \tau \quad (16)$$

Action Potential Integral

$$\begin{aligned} A &= -m \int_a^b \Delta \tau \\ &= -m \int_a^b \sqrt{dt^2 - dx^2} \\ &= -m \int_a^b \sqrt{1 - \frac{dx^2}{dt^2}} dt \\ &= -m \int_a^b \sqrt{1 - \dot{x}^2} dt \end{aligned} \quad (17)$$

Lagrangian

$$\begin{aligned} \mathcal{L} &= -m \sqrt{1 - (\dot{x}^2 + \dot{y}^2 + \dot{z}^2)} \\ &= -m \sqrt{1 - v^2} \end{aligned} \quad (18)$$

Lagrangian's Lagrangian

$$\mathcal{L}_{\mathcal{L}} = \frac{m\dot{x}^2}{2} \quad (19)$$

where

$$\dot{x}^2 = \sum \dot{x}_i^2 \quad (20)$$

Continue the Action potential integral and get Newtonian Energy

$$\begin{aligned} A &= -m \int_a^b \mathcal{L} \\ &= -m \left(1 - \frac{v^2}{2} \right) \\ &= mv^2 \end{aligned} \quad (21)$$

ignoring the constant $-m$ from the final value.

3 Conservation of Energy and Momentum

Lagrangian momentum

$$\begin{aligned} P_x &= \frac{\partial \mathcal{L}}{\partial \dot{x}} \\ &= \frac{m\dot{x}}{2\sqrt{1-v^2}} \\ &= \frac{mv_x}{\sqrt{1-v^2}} \end{aligned} \quad (22)$$

where

$$v^2 = x^2 + y^2 + z^2 \quad (23)$$

This looks like U_x , hence, relativistic momentum uses the mass at rest and the relativistic space axes x, y, z . e.g. x :

$$P_x = \frac{mv_x}{\sqrt{1-v^2}} \quad (24)$$

4-vector in terms of deltas $\Delta t, \Delta x$.

Momentum by component

$$\begin{aligned} mU^x &= P_x \\ mU^y &= P_y \\ mU^z &= P_z \\ mU^0 &= P_0 \end{aligned} \quad (25)$$

Conservation of momenta

$$\vec{P}_{initial} = \vec{P}_{final} \quad (26)$$

or

$$\vec{P}_{initial} - \vec{P}_{final} = 0 \quad (27)$$

What is P_0 ? Is it the particle energy? Look at the Hamiltonian.

$$H = \sum_i \dot{X}_i P^i - \mathcal{L} \quad (28)$$

By component, P^i

$$P^i = \frac{m\dot{X}^i}{\sqrt{1-v^2}} \quad (29)$$

By component, $\mathcal{L}$

$$\mathcal{L} = -m\sqrt{1-v^2} \quad (30)$$

Rewriting the Hamiltonian

$$H = \sum_i \frac{m\dot{x}_i^2}{\sqrt{1-v^2}} + m\sqrt{1-v^2} \quad (31)$$

using

$$v^2 = \sum_i \dot{x}_i^2 \quad (32)$$

we get and simplify

$$\begin{aligned} H &= \frac{mv^2}{\sqrt{1-v^2}} + m\sqrt{1-v^2} \\ &= \frac{mv^2}{\sqrt{1-v^2}} + \frac{m(\sqrt{1-v^2})^2}{\sqrt{1-v^2}} \\ &= \frac{mv^2 + m - mv^2}{\sqrt{1-v^2}} \\ &= \frac{m}{\sqrt{1-v^2}} = mU_0 \end{aligned} \quad (33)$$

To calculate energy of a particle at rest, we use the Taylor expansion of

$$\frac{1}{\sqrt{1-v^2}} = (1-v^2)^{-\frac{1}{2}} \approx 1 + \frac{v^2}{2} \quad (34)$$

And get energy

$$H = m + \frac{mv^2}{2} \quad (35)$$

Note that at rest, velocity is zero so energy is simply m , however, the units of H , to be consistent with $\frac{mv^2}{2}$ must be $\frac{gm^2}{s^2}$ hence $\frac{mv^2}{2}$ drops and fixing the units, rest energy is:

$$H_{rest} = mc^2 \quad (36)$$

4 Massless Particles

Clearly, $m = 0$, $v = c$ causes energy, $\frac{m}{1-v^2}$, to go to $\frac{0}{0}$ so instead we use momentum. Recall

$$U^{0^2} - U^{x^2} - U^{y^2} - U^{z^2} = 1 \quad (37)$$

Multiply by m

$$m^2 U^{0^2} - m^2 U^{x^2} - m^2 U^{y^2} - m^2 U^{z^2} = m^2 \quad (38)$$

Substitute energy and momentum

$$E^2 - P_x^2 - P_y^2 - P_z^2 = m^2 \quad (39)$$

To Energy:

$$E = \sqrt{P^2 + m^2} \quad (40)$$

With units

$$E = \sqrt{P^2 c^2 + m^2 c^4} \quad (41)$$

As massless:

$$E = c|P| \quad (42)$$

5 positronium

An electron and an positron in orbit, and will decay into two photons, leaving in opposite directions. Calculate the energy and momentum of the system before and after decay. Initial mass in joules or grams. Momentum is conserved. At rest, zero. Photons must go in opposite directions for momentum to be zero. And equal momentum, to ensure zero momentum.

Start with

$$E_{rest} = m_{positronium} c^2 \quad (43)$$

This must equal the end state, twice the momenta of each photon:

$$mc^2 = 2c|P| \quad (44)$$

Rearrange

$$\frac{mc}{2} = |P| \quad (45)$$

6 Energy with Units

$$E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (46)$$

7 Lecture 4

Fields. Discussion of field function from t and x to ϕ .

$$\phi(t, x) \quad (47)$$

The action

$$Action = \int_a^b \frac{m}{2} \left(\frac{d\phi}{dt} \right)^2 - V(\phi) dt \quad (48)$$

Lagrangian

$$\mathcal{L} = \frac{m}{2} \dot{\phi}^2 - V(\phi) \quad (49)$$

Equations of motion given by

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \frac{\partial \mathcal{L}}{\partial \phi} \quad (50)$$

Carry out computation to get Newton's equations:

$$m\ddot{\phi} = -\frac{\partial V}{\partial \phi} \quad (51)$$

Discussion of least action from point a to point b . Cut time axis into slices, the integral is a sum of the value of ϕ at each slice. To minimize, take the derivative.

We are *Three plus one* for three space dimensions and one time dimension. Hold fixed the boundary, wiggle up and down, until we minimize the action. In practice, the Action Potential is

$$A = \int \int \int \int dx dy dz dt \quad (52)$$

But in shorthand

$$A = \int d^4x \mathcal{L} \left(\phi, \frac{\partial \phi}{\partial t}, \frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial z} \right) \quad (53)$$

Condense once

$$A = \int d^4x \mathcal{L} \left(\phi, \frac{\partial \phi}{\partial x^\mu} \right) \quad (54)$$

And use

$$\begin{aligned} \phi_t &= \frac{\partial \phi}{\partial t} \\ \phi_x &= \frac{\partial \phi}{\partial x} \\ \phi_y &= \frac{\partial \phi}{\partial y} \\ \phi_z &= \frac{\partial \phi}{\partial z} \end{aligned} \quad (55)$$

To condense notation further, rather than use $\dot{\phi}$

How to handle a field, which is a surface? Some other notes describe the Euler Lagrange equations, but here we have:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \quad (56)$$

Or starting here:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \frac{d\phi}{dt}} - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \quad (57)$$